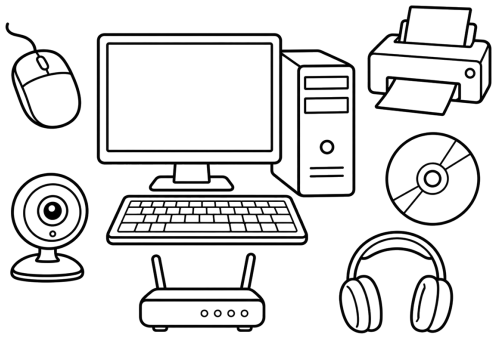
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 10



UNIT 2
Fundamentals of a Computer System

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain what a computer is and describe its five main features.
- [ ] Classify computers by their physical size and by the technology they use.
- [ ] Explain how data and control signals move in a computer system.
- [ ] Name input and output devices, and the three parts of the CPU.
- [ ] Compare RAM, ROM and cache, and magnetic, optical and solid state storage.
- [ ] Identify the ports of a computer and the devices that connect to each.
- [ ] Describe data communication, network devices, network types and topologies.

THREE RULES FOR THE COMPUTER ROOM

- **Always** ask your teacher before you unplug a cable or open a computer.
- **Never** touch the inside of a computer while it is connected to the mains.
- **Always** put devices, cables and discs back in their places after the lesson.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical task: look at real devices, ports and the network of your computer lab. Last you **write** the exercises in this book.

You will need: a pencil and an eraser, your textbook (pages 37 - 72), and the computers, devices and network of your school computer lab.

THE WORDS YOU WILL LEARN

Computer	Versatility	Super computer	Server
Hybrid computer	Input device	Output device	Soft copy
Hard copy	CPU	ALU	RAM
ROM	Cache memory	Secondary storage	Port
Simplex	Full duplex	Router	Modem
Firewall	LAN	WAN	Topology

"A computer is only a machine. You are the one who gives it a purpose."

- THE KING OF ICT

CHAPTER 1

The Computer and Its Classification

? A NEW COMPUTER LAB

"Our school is building a new computer lab. Should we buy a super computer?"

"No! Super computers are for NASA. We need micro computers - and a server."

Computers come in many sizes and kinds. In this chapter you learn what a computer is, its **features**, the **types** of computers and how a **computer system** works.

2.1 What is a computer?

A computer is an **electronic device** which accepts or collects **data**, **processes** them according to the given **instructions** and produces the desired **output**.

Feature	What it means
Speed	It does a calculation in a few seconds. It executes more than one million instructions per second.
Accuracy	It gives the correct output when the correct instructions and data are given.
Efficiency	It never gets tired. It can work round the clock with the same level of accuracy.
Versatility	It can be used to do many tasks at the same time.
Storing and retrieving	It stores a large amount of data in a small unit, and finds it again easily and quickly when needed.

2.2 Classification by physical size

Computers can be classified according to their **physical size** and according to the **technology** they use (Figure 2.1).

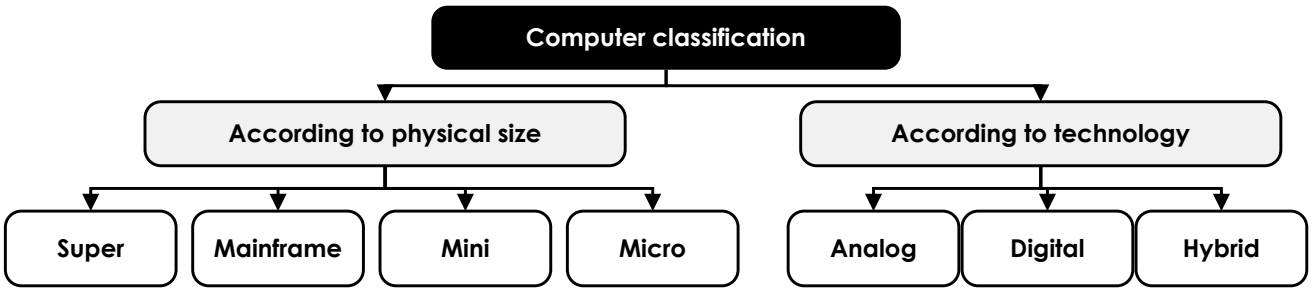


Figure 2.1 - computer types

"A computer never gets tired. Neither should your will to learn."

- THE KING OF ICT

Type	What it is like	Where it is used
Super computers	The highest computing power (the tasks done in a given time). Large, expensive, rare.	Science and engineering: NASA, large businesses, the military.
Mainframe computers	Less size, power and cost than super computers. Many users share one computer through **terminals**.	Large scale businesses and e-business.
Mini computers	Smaller, less powerful and cheaper than mainframes. A few users share it through terminals. Also called common purpose computers.	Medium scale institutions such as banks.
Micro computers	Small computers for personal use: small memory, less speed, low cost, little electricity.	Desktops, laptops, hand-held computers, smart phones.

The words "mainframe" and "mini" are going out of use: these computers are now called **server computers**.

2.3 Classification by technology

Type	What it uses	Examples
Analog computers	Use **analog signals**: environmental quantities such as speed, pressure and temperature.	Speedometers, road lamps with sensors, meteorological machines.
Digital computers	Use **digital signals**. The computers we use every day.	Desktop computers, laptops, smart phones.
Hybrid computers	A **combination** of analog and digital computers.	The ECG machine: it reads the heart beat (analog), changes it to a digital signal and prints it.

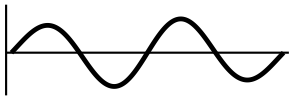


Figure 2.8 - analog signal

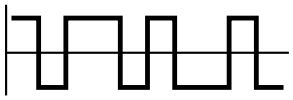


Figure 2.9 - digital signal

2.4 The function of a computer system

A computer system **inputs**, **processes** and **stores** data, and produces **information**.

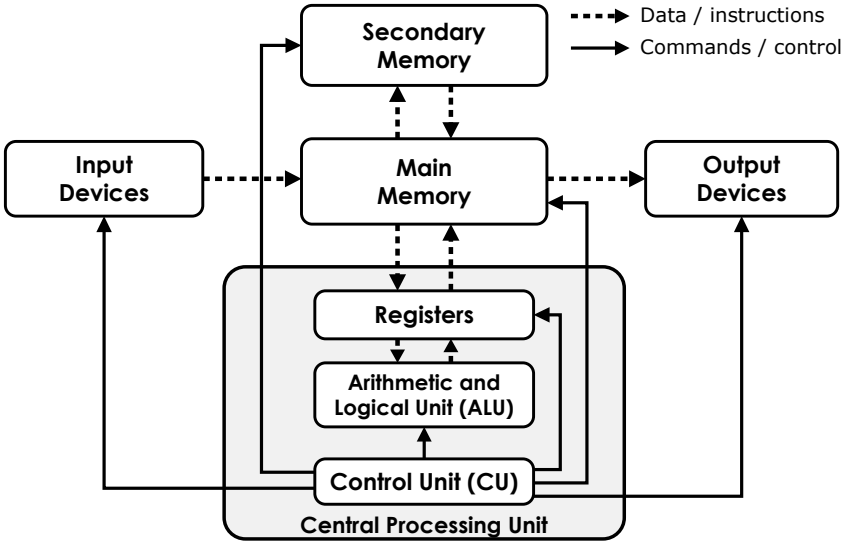


Figure 2.10 - a computer system

- Data and instructions from the **input devices** go to the **main memory**, and then to the **CPU** for processing.
- Information from the main memory goes to the **output devices**.
- Data to keep goes to the **secondary memory**.
- The **Control Unit** sends control signals to all the devices.
- **Dotted** lines: data and instructions. **Solid** lines: control signals.

A Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

processes	million	versatility	terminals
server	micro	temperature	Control Unit

1. A computer accepts data, ... them according to the given instructions and produces the desired output.
2. A computer executes more than one ..... instructions per second.
3. Doing many tasks at the same time is called ..... .
4. In a mainframe, many users share one computer through ..... .
5. Mainframe and mini computers are now called ..... computers.
6. Laptops, hand-held computers and smart phones are ..... computers.
7. Analog computers use quantities such as speed, pressure and ..... .
8. In a computer system, the ..... sends control signals to all the devices.

B Analog, digital or hybrid?

6 marks

Tick one box for each computer.

The computer	Analog	Digital	Hybrid
A speedometer in a vehicle	☐	☐	☐
An ECG machine in a hospital	☐	☐	☐
The laptop you use at home	☐	☐	☐
A meteorological (weather) machine	☐	☐	☐
A road lamp that switches on with a sensor	☐	☐	☐
A smart phone	☐	☐	☐

C True or false

6 marks

Write T or F in the box.

1.	☐	A computer gives correct output only when correct instructions and data are given.
2.	☐	Super computers are small, cheap and common.
3.	☐	Mini computers are more powerful than mainframe computers.
4.	☐	Mainframe and mini computers are now called server computers.
5.	☐	In Figure 2.10, control signals are shown by a dotted line.
6.	☐	Data from an input device first goes to the primary memory.

D Match the computers

6 marks

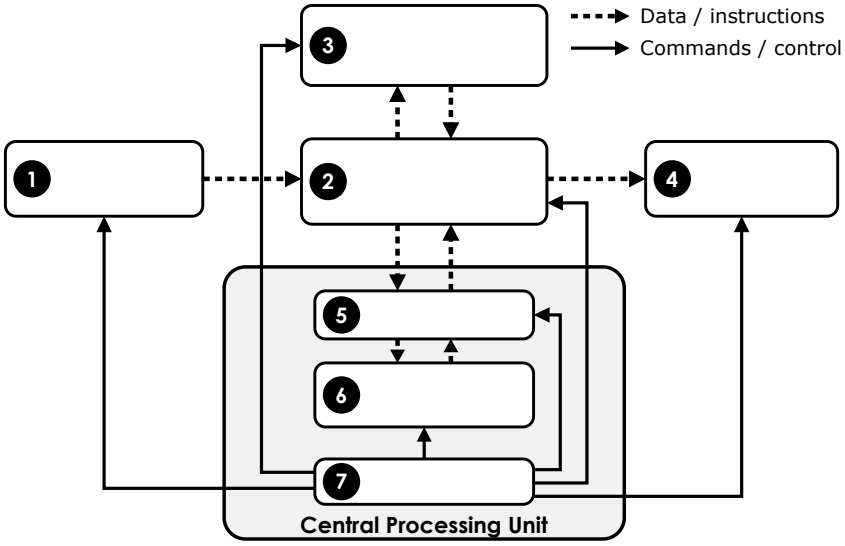
Write the correct letter in the box.

A - The computer		B - What it is like
1. Super computer		**A.** A common purpose computer used in medium scale institutions such as banks.
2. Mainframe computer		**B.** A combination of analog and digital, like the ECG machine.
3. Mini computer		**C.** The highest computing power; used by NASA and the military.
4. Micro computer		**D.** For personal use, like a laptop or a smart phone.
5. Analog computer		**E.** It uses signals such as speed, pressure and temperature.
6. Hybrid computer		**F.** Many users share it through terminals in large businesses.

E Complete Figure 2.10

7 marks

Write the names in the numbered boxes.



USE THESE WORDS

Input devices
Output devices
Main memory
Secondary memory
Registers
ALU
Control Unit

CHAPTER 2

Input Devices, Output Devices and the CPU

? A NOISY PRINTER

"The lab printer is so noisy! The office printer is quiet and its pages are much clearer."

"The lab has a dot matrix printer. The office has a laser printer."

A computer needs **input devices** to get data and **output devices** to give information. In between, the **CPU** does the work.

2.5 Input devices

Data and instructions are fed to the computer using **input devices**. The **keyboard** is the most common one. Its keys are in five groups (Figure 2.11).

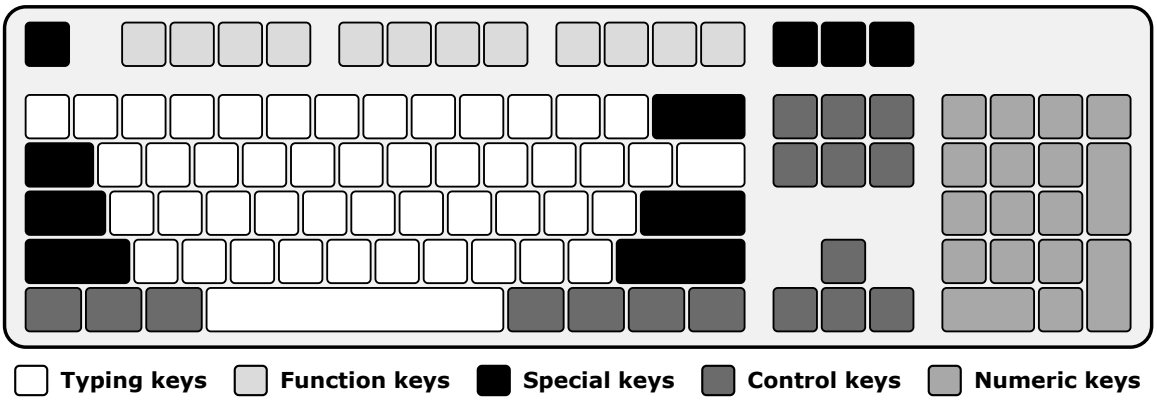


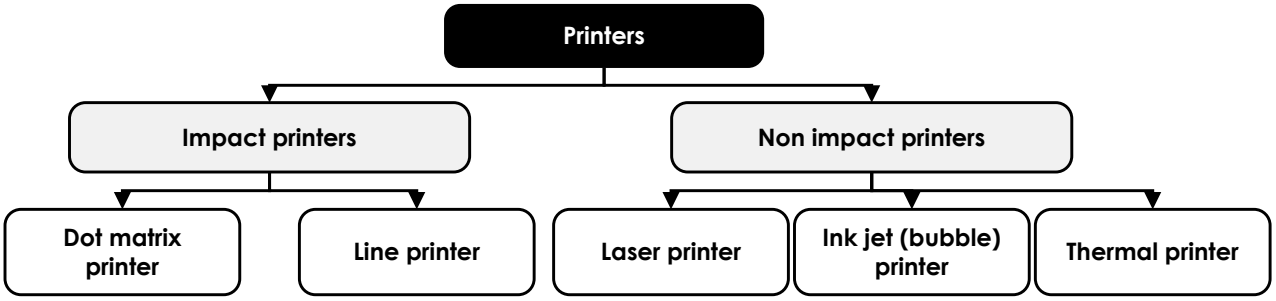
Figure 2.11 - the keyboard

Group	Device	What it does
Keyboard	Keyboard	The most common input device: 101/102 or 104/108 keys.
Pointing devices	Mouse	The most popular pointing device: two buttons, a wheel.
	Touch pad	The pointing device of a laptop computer.
	Touch screen	Data is fed by touching it. It is input AND output.
	Joystick	Moves the cursor in computer games.
	Light pen	Selects or draws on the screen. Used in CAD.
Imaging and video	Digital camera	Inputs photographs and videos.
	Webcam	Shows a user over the Internet; captures videos.
	CCTV	Pictures for road safety and the security of places.
Scanners and readers	Flatbed scanner	Like a photocopier: the image becomes digital data.
	Barcode reader	Changes a bar code into an alphanumeric value.
	MICR	Magnetic Ink Character Reader: numbers on a cheque.
	OCR	Optical Character Recognition: captures text to edit.
	OMR	Optical Mark Recognition: pen or pencil marks (MCQs).
	ATM	Reads a bank card and connects to the bank computer.
Sound	Microphone	Feeds sound to the computer.

2.6 Output devices

Output comes as a **soft copy** (on the screen), a **hard copy** (tangible, such as a printout) or **sound** (speakers and headphones).

Soft copy	What it is
CRT	Cathode Ray Tube monitor: large; uses more electricity.
LCD	Liquid Crystal Display: a flat screen with liquid crystal between two polarising sheets.
LED	Light Emitting Diode screen: brighter, low electricity; billboards, name boards.
Multimedia projector	Shows the output on a wide screen for many people.



The two main types of printers (textbook page 48)

Impact printers strike an ink ribbon with a head or needles: noisy, but the ribbon is cheap, so banks and shops use them for receipts. **Non impact printers** are quiet and print quickly and better, but cost more to run.

Hard copy	Kind	How it works
Dot matrix	Impact	Needle-like tips strike an inked ribbon and print small dots.
Line printer	Impact	Prints one row at a time: about 3000 lines per minute.
Laser printer	Non impact	Toner (dried ink) from a cartridge goes on the paper. Clear, quality prints.
Ink jet (bubble)	Non impact	Liquid ink goes onto the paper through small nozzles.
Thermal printer	Non impact	Uses heat. Receipts in ATMs and shops, labels.
Plotter	Hard copy	Draws plans with a moving pen: architecture, town planning.

2.7 The Central Processing Unit (CPU)

The CPU is the main component of a computer system. It is called the **brain** of the computer. It executes instructions as directed by the operating system. It has three main parts:

Part of the CPU	What it does
Arithmetic and Logical Unit (ALU)	Does all the mathematical and logical functions.
Control Unit (CU)	Controls all the devices of the computer system.
Memory registers	Temporarily store the data needed by the ALU and the information it produces.

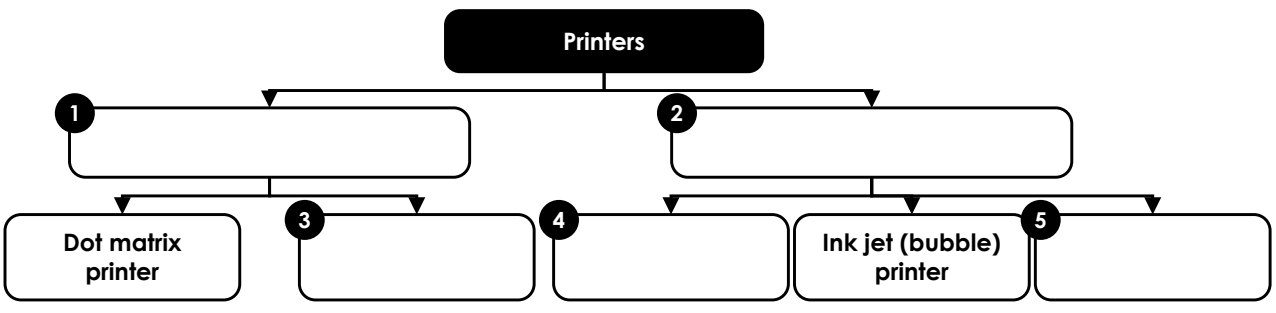
(i) COMPUTER SPEED

The clock speed of a computer is measured in **Hertz (Hz)**. Computer speeds are given in **megahertz (MHz)** and **gigahertz (GHz)**.

A Complete the printer tree

5 marks

Write the missing names in the numbered boxes.



B Input, output or both?

8 marks

Tick one box for each device.

The device	Input	Output	Both
Light pen	☐	☐	☐
Plotter	☐	☐	☐
Touch screen	☐	☐	☐
MICR	☐	☐	☐
Multimedia projector	☐	☐	☐
Webcam	☐	☐	☐
Thermal printer	☐	☐	☐
OMR device	☐	☐	☐

C Which device would you use?

8 marks

Write the name of one device for each job.

You want to ...	Device
Read the numbers at the bottom of a cheque.	
Mark the MCQ answer sheets of an exam.	
Capture the text of an old document so that you can edit it.	
Read the code on a packet of biscuits at the supermarket.	
Draw on the screen in a Computer Aided Design program.	
Print the plan of a new house for an architect.	
Print a receipt in an ATM.	
Show a lesson on a wide screen to the whole class.	

D Impact or non impact?

6 marks

Tick one box for each printer.

The printer ...	Impact	Non impact
A head or a needle strikes an ink ribbon.	☐	☐
It is not noisy.	☐	☐
Its ribbon is cheap, so banks and shops use it for receipts.	☐	☐
Toner (dried ink) from a cartridge is put on the paper.	☐	☐
It prints about 3000 lines per minute.	☐	☐
Heat is used to print the document.	☐	☐

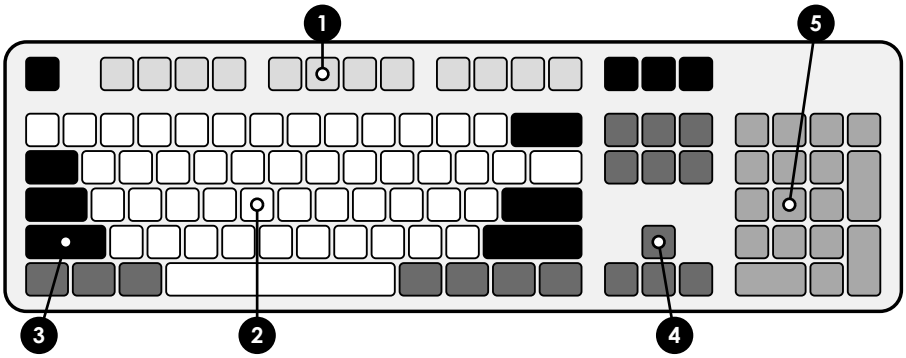
"Input good habits every day, and your life will output success."

- THE KING OF ICT

E Label the keyboard

5 marks

Write the name of each group of keys (Figure 2.11).



1. .....
2. .....
3. .....
4. .....
5. .....

F Match them

6 marks

Write the correct letter in the box.

A - Device or part		B - What it is
1. CRT monitor		**A.** Does all the mathematical and logical functions.
2. LCD monitor		**B.** Large in size and uses more electricity.
3. LED screen		**C.** Temporarily store the data for the ALU and its results.
4. ALU		**D.** Liquid crystal between two sheets of polarising material.
5. Control Unit		**E.** Controls all the devices of the computer system.
6. Memory registers		**F.** Better brightness; used on billboards and name boards.

G Choose the correct answer

6 marks

Underline your answer.

1. Which units are used to measure computer speed?
   (a) bytes and bits (b) MHz and GHz
   (c) pixels and dots (d) metres and miles

2. The output that we see on the screen is called
   (a) a soft copy (b) a hard copy (c) a printout (d) a backup

3. Which device is both an input and an output device?
   (a) Plotter (b) Scanner (c) Touch screen (d) Microphone

4. The CPU is called the ... of the computer.
   (a) heart (b) eye (c) brain (d) hand

5. Which printer prints one row at a time?
   (a) Laser printer (b) Line printer (c) Thermal printer (d) Ink jet printer

6. A keyboard usually has
   (a) 26 to 30 keys (b) 50 to 60 keys (c) 101 to 108 keys (d) 200 to 250 keys

CHAPTER 3

Memory, Storage and Ports

? WHERE DID MY ESSAY GO?

"The power went off and I lost the essay I was typing! But my photos on the hard disk are safe. Why?"

Your essay was only in **RAM**, which forgets everything when the power goes off. The photos were saved on **secondary storage**. In this chapter you learn about memory, storage and the **ports** that connect devices.

2.8 Computer memory

Memory and storage devices store data and information. The memory that the CPU can reach directly is the **primary memory**. There are three types of primary memory.

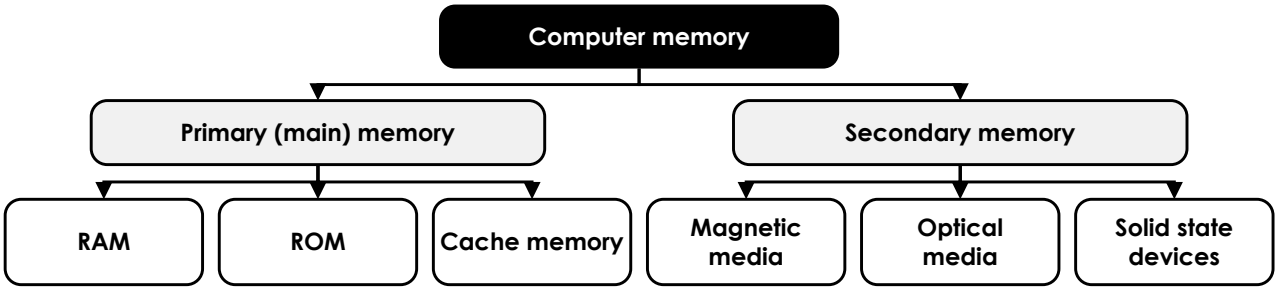


Figure 2.39 - computer memory

Primary memory	What it is
RAM	Random Access Memory, often called the **main memory**. Stores data **temporarily**: the data is deleted when the computer is switched off, so it is **volatile**. Holds the data from input devices and to output devices. It can be read and written.
ROM	Read Only Memory. Its data is **not erased** when the computer is switched off: **non-volatile**. It keeps the commands to boot the computer, the **BIOS** (Basic Input Output System). A small ROM is fixed to the motherboard by the manufacturer.
Cache memory	Smaller than other memory but **faster**; also called **CPU memory**. It is the mediator between the CPU and the primary memory and keeps the data used most often.

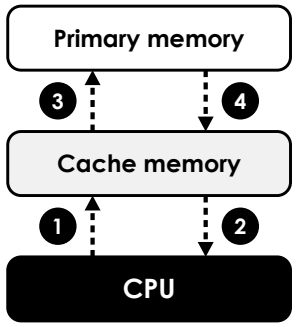


Figure 2.42 - cache memory

1. The CPU checks the cache memory for the data it needs.
2. If the data is there, it is fetched from the cache.
3. If not, the CPU checks the primary memory (RAM).
4. The data is fetched from the primary memory to the cache, and then sent to the CPU.

2.9 Secondary storage

Secondary memory (secondary storage, also called external storage) keeps data and information **permanently**. It is **non-volatile**: it does not lose data when there is no electricity.

Type	How it works	Examples
Magnetic media	Data is stored using a **magnetic field**.	Hard disk, floppy disk, magnetic tape
Optical media	Data is read and written by a **laser beam**.	CD, DVD, Blu-ray disc
Solid state devices	**No moving parts**. They read and write data.	USB flash drive, memory card

The hard disk - The hard disk drive (HDD) is inside the computer; there are external hard disks too. You can read and write its data, so it is called **read / write memory**. Capacity: from **250 GB to 4 TB**, and it grows with technology.

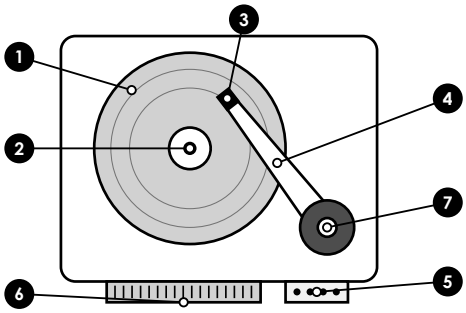


Figure 2.44 - the hard disk

1. **Platter** - a metal or ceramic disk; data on both sides
2. **Spindle** - holds the platter and rotates it
3. **Head** - reads and writes the data (two for a platter)
4. **Actuator arm** - moves the head over the platter
5. **Power connector** - connects the hard disk to power
6. **IDE connector** - transmits the data
7. **Actuator** - a motor-driven device that moves the arm

Optical disc	Capacity	Kinds
CD	650 - 900 MB	CD-ROM, CD-R, CD-RW
DVD	4.7 - 9.4 GB	DVD-ROM, DVD-R, DVD-RW
Blu-ray disc (BD)	25 - 128 GB	BD-R, BD-RE, Blu-ray 3D

Names ending in **ROM** = read only: you can only read the data written on it; **R** = recordable: data can be written once, until it is full; **RW / RE** = re-writable: data can be erased and written again.

Magnetic tape is used in server computers for backup because of its large capacity, but it is slow. **Solid state** devices: the **USB flash drive** is easy to carry and fast; a **memory card** (for example an SD card - Secure Digital) is smaller and is used in digital cameras, mobile phones and gaming devices.

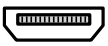
DO IT - PRACTICAL 1 (TEXTBOOK ACTIVITY, PAGE 55)

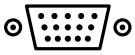
Storage media and the parts of a hard disk. **Only your teacher opens a hard disk.**

1	In groups, write the advantages and disadvantages of each type of storage media.
2	Compare magnetic, optical and solid state media: capacity, speed, price, carrying.
3	Your teacher shows an opened (discarded) hard disk. Do not touch the platter.
4	Find the seven parts of Figure 2.44 on the real hard disk.

2.10 Computer ports

Ports are the interfaces that connect a computer with its devices or with other computers. They are at the front or the back of the computer, and all of them are connected to the **motherboard**. HDMI = High-Definition Multimedia Interface; USB = Universal Serial Bus; VGA = Video Graphics Adaptor.

Port	Name and use
	**PS/2 port** - Keyboard (purple) and mouse (green). Today USB is used more.
	**Parallel port** - Printer. It has 25 holes. Today USB is used more.
	**HDMI port** - Monitor, multimedia projector, digital TV, digital speakers.
	**Ethernet port (RJ45)** - Joins the computer to a network router or switch.

Port	Name and use
	**Audio ports / jacks** - Blue: line in. Green: line out, headphones. Pink: microphone.
	**USB port** - Many kinds of input and output devices.
	**VGA video port** - Monitor or multimedia projector. 15 holes. Also called the graphic port.
	**Serial port** - Devices such as a modem. Today USB is used instead.



USB type A
Keyboards, mice, flash drives and printers.



USB type B
Square shape. Printers and external hard disks.



USB type C
Oval; fits either way round. Fast charging and data: phones, laptops.

DO IT - PRACTICAL 2 (TEXTBOOK ACTIVITY, PAGE 58)

Observe and identify the ports of a computer in the laboratory.

1	Shut down a lab computer. Your teacher switches it off at the mains.
2	Look at the back and the front of the system unit (and the sides of a laptop).
3	Find each port of section 2.10. Count the USB ports. Look at the jack colours.
4	Draw the back panel and label the ports. Do not unplug any cable.

A Magnetic, optical or solid state?

8 marks

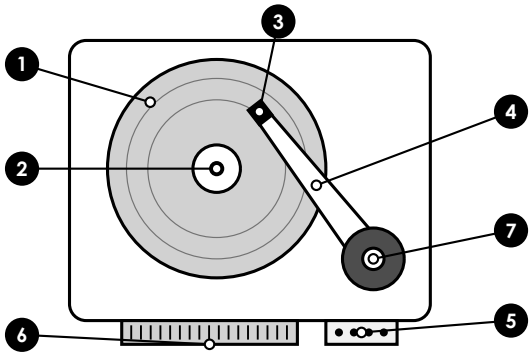
Tick one box for each device.

The device	Magnetic	Optical	Solid state
Hard disk	☐	☐	☐
CD-RW	☐	☐	☐
USB flash drive	☐	☐	☐
Magnetic tape	☐	☐	☐
DVD	☐	☐	☐
Memory card (SD card)	☐	☐	☐
Blu-ray disc	☐	☐	☐
Floppy disk	☐	☐	☐

B Label the hard disk

7 marks

Write the names of the parts 1 - 7 (Figure 2.44).



WORD BANK

Actuator	Actuator arm
Head	IDE connector
Platter	Power connector
Spindle	

1. .....
2. .....
3. .....
4. .....
5. .....
6. .....
7. .....

C RAM, ROM or cache?

6 marks

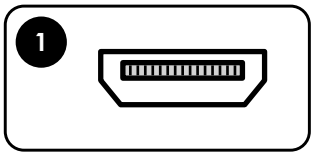
Tick one box for each sentence.

The memory ...	RAM	ROM	Cache
Its data is lost when the power is switched off.	☐	☐	☐
It keeps the BIOS (the commands to boot).	☐	☐	☐
It is between the CPU and the primary memory.	☐	☐	☐
It is often called the main memory.	☐	☐	☐
It is small but fast; also called CPU memory.	☐	☐	☐
It is non-volatile and fixed to the motherboard.	☐	☐	☐

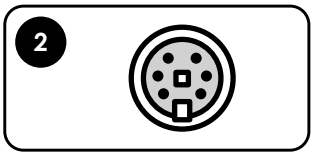
D Name the ports

8 marks

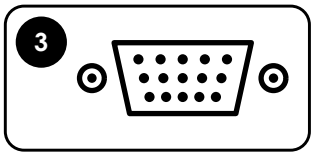
Write the name of each port on the line under it.



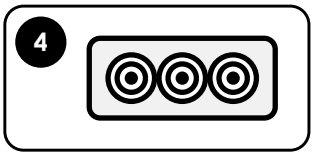
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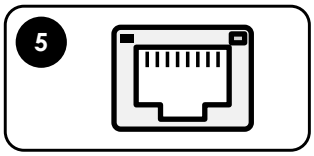
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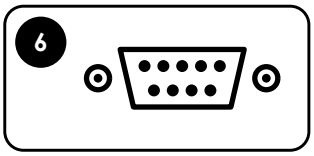
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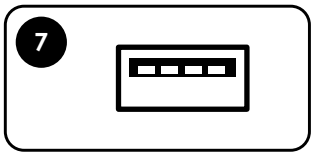
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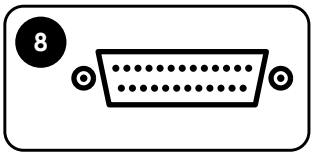
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E Which port would you use?

6 marks

Write the name of the port.

You want to ...	Port
Connect a multimedia projector to a laptop.	
Join a computer to the network switch.	
Plug in a microphone.	
Connect an old printer that has a 25-pin plug.	
Connect an old modem.	
Connect a mouse that has a round green plug.	

F Complete the tables

6 marks

Write the capacity of each disc, and what each name means.

Disc	Capacity
CD	
DVD	
Blu-ray disc	

Disc	What it means
CD-ROM	
DVD-R	
CD-RW	

CHAPTER 4

Data Communication and Computer Networks

? ONE SECOND TO JAFFNA

"I sent a photo to my cousin in Jaffna and she got it in one second! How did it travel?"

Data travels from a **sender** to a **receiver** through a **medium**: cables or waves. Devices such as switches and routers join computers into **networks**.

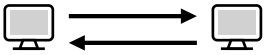
2.11 Data communication

Transmitting data and information is called **data communication**. It can be between two or more computers, or devices such as printers. It needs a connection between a sender and a receiver; creating such a connection is called **networking**. The basic components are:



Figure 2.59 - the components of data communication

2.12 Data transmission modes

Mode	Picture	What happens	Examples
Simplex		Data goes from the sender to the receiver **only**.	Getting printouts, watching television, listening to the radio
Half-duplex		Data goes in **one direction at a time**. After the sender sends, the receiver can send back.	Browsing the Internet, a walkie-talkie
Full duplex		Data goes in **both directions at the same time**.	A telephone conversation

2.13 Data transmission media

There are two types of data transmission media (Figure 2.63). A medium is **guided (wired)** when it is a physical cable, and **unguided (wireless)** when the data travels as signals through the air.

Type	Medium	What it is like	Used for
Guided (wired)	Twisted pair: UTP	Unshielded twisted copper pairs. Flexible and cheap; up to about 100 m.	Telephone lines
	Twisted pair: STP	Shielded twisted pairs: better quality and more secure, but expensive.	Secure data links
	Coaxial cable	A central cable inside a braided copper net, with plastic between. Expensive.	TV antennas, CCTV
	Fibre optics	A glass core in glass cladding; data travels as reflected light. The **fastest** medium, but expensive.	Modern telephone networks
Unguided (wireless)	Radio waves	Data goes through the air as radio signals.	Wi-Fi, Bluetooth
	Microwaves	Travel in a straight line. Satellites above 36000 km send them back to Earth, so data can go any distance.	Satellites, the Internet
	Infrared	Wireless signals over a short distance.	TV remotes, wireless keyboards

DO IT - PRACTICAL 3 (TEXTBOOK ACTIVITY, PAGE 62)

Copper wire or optical fibre? A group discussion.

1	In groups, list the advantages and disadvantages of copper wire and optical fibre.
2	Compare them for speed, distance and price.
3	Choose the best medium for: a school lab, a link between two cities, a TV remote.
4	Share your choices with the class and give your reasons.

"Alone we can do a little. Connected, like a network, we can do so much more."

- THE KING OF ICT

2.14 Computer network devices

A network needs two or more computers and devices to connect them:

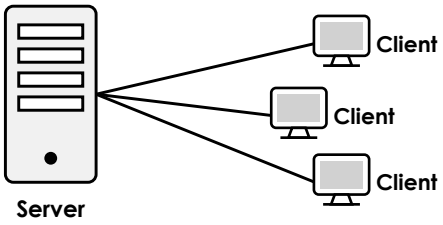
Device	What it does
NIC	Network Interface Card. It has an RJ45 port; built into the motherboard of modern computers.
Switch / Hub	Join computers with cables. A **hub** sends data to **all** the computers (half duplex; it can congest the network). A **switch** sends data only to the **relevant** computer (full duplex).
Wi-Fi	Wireless Fidelity: access points join computers, phones and laptops with radio waves.
Router	Connects two or more networks; also called a **gateway**. Finds the best path for data.
Modem	Changes digital signals to analog for the telephone line, and back: **MO**dulation and **DEM**odulation. Internal, external or wireless.
Firewall	A secured barrier between the internet and a network. Software or hardware.

DO IT - PRACTICAL 4 (TEXTBOOK ACTIVITY, PAGE 65)

Observe the firewall of the operating system with the help of your teacher.

1	With your teacher, open the **Control Panel** and click **System and Security**.
2	Click **Windows Defender Firewall**. Is it on for private and public networks?
3	Open **Allow an app or feature through Windows Defender Firewall**. Change nothing.
4	Write in your book what a firewall does.

2.15 Client/server and the types of networks



A **server** computer responds to the **client** computers connected to it. A network operating system is installed on the server. The one who requests data or information is the client; the one who provides it is the server. Networks are classified by how far apart their computers are.

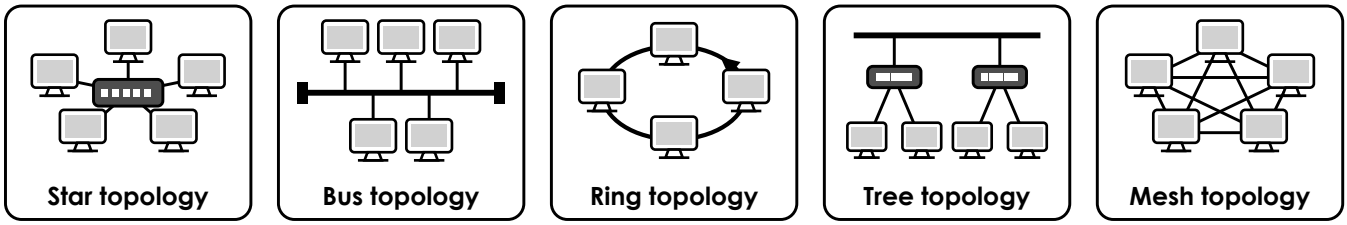
Type	Name	How far it reaches	Examples
LAN	Local Area Network	A room or several rooms in a building.	A school, a university, a small business, an office
PAN	Personal Area Network	The smallest: about 1 to 10 metres. Bluetooth, Wi-Fi, infrared, USB.	A phone, laptop, tablet, earphones, printer
MAN	Metropolitan Area Network	Several LANs joined; less than 16 miles.	Banking networks, universities with branches
WAN	Wide Area Network	A wide area, inside or outside a country.	The Internet

"Share what you know. Knowledge grows when it travels across the network."

- THE KING OF ICT

2.16 Network topology

Network topology is the pattern of connection in designing a computer network.



Topology	How it is built
Star topology	Built around a central **switch / hub**. Commonly used.
Bus topology	All computers on one main cable, the **backbone**. Easy and fewer cables, but only a limited number of computers; a backbone problem affects all.
Ring topology	Computers in a ring. Data moves in a **circle**. One broken computer or cable can stop the whole network.
Tree topology	Several star networks joined to a bus backbone. Each part can be controlled on its own.
Mesh topology	Every computer is joined to every other. Costly and hard to control, but one breakdown does not stop the network. Found on the internet.

2.17 Benefits and disadvantages of networking

★ BENEFITS OF NETWORKING

- Share data and information.
- Less storage space: data is not repeated.
- Share printers, scanners and modems.
- Share software installed on a server.
- Reach your work anywhere, any time.
- Security: only authorised clients.
- Send e-mail.

! DISADVANTAGES

- Security issues: hackers may attack.
- A broken cable or computer may stop others.
- A virus can spread to other computers.
- A server breakdown affects the clients.
- A trained administrator is needed.

A Match the network devices

6 marks

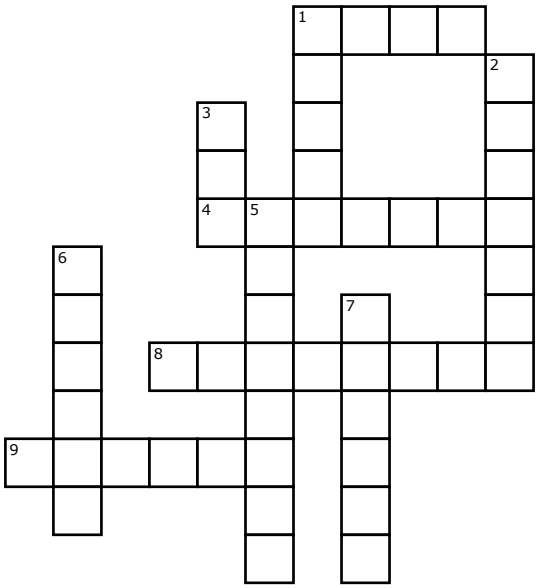
Write the correct letter in the box.

A - The device		B - What it does
1. NIC		**A.** Changes digital signals to analog signals and back.
2. Hub		**B.** Sends the data only to the computer it is meant for.
3. Switch		**C.** Has an RJ45 port; built into the motherboard of modern computers.
4. Router		**D.** A secured barrier between the internet and a network.
5. Modem		**E.** Sends the data to all the computers of the network.
6. Firewall		**F.** Connects two or more networks; also called a gateway.

B Crossword

10 marks

All the words are in this chapter. Write one letter in each square.



ACROSS

1. Every computer is joined to every other computer (4)
4. Data goes from the sender to the receiver only (7)
8. A barrier between the internet and a network (8)
9. It provides data and information to its clients (6)

DOWN

1. It changes digital signals to analog signals and back (5)
2. A cable used for TV antennas and CCTV (7)
3. A topology with one backbone cable (3)
5. TV remote controllers use it (8)
6. It connects two or more networks (6)
7. It sends data only to the relevant computer (6)

C Simplex, half-duplex or full duplex?

6 marks

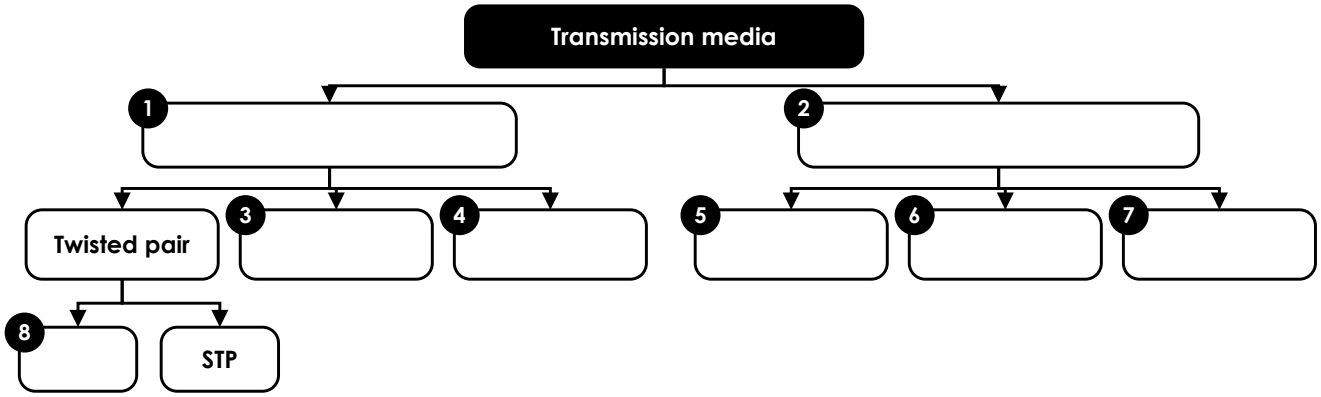
Tick one box for each.

The communication	Simplex	Half-duplex	Full duplex
Watching television	☐	☐	☐
Talking on the telephone	☐	☐	☐
A walkie-talkie used by security forces	☐	☐	☐
Listening to the radio	☐	☐	☐
Browsing the Internet to get information	☐	☐	☐
Getting a printout	☐	☐	☐

D Complete the media tree

8 marks

Write the missing names in the numbered boxes (Figure 2.63).



"Build a firewall around your study time. Keep out everything that wastes it."

- THE KING OF ICT

E LAN, PAN, MAN or WAN?

6 marks

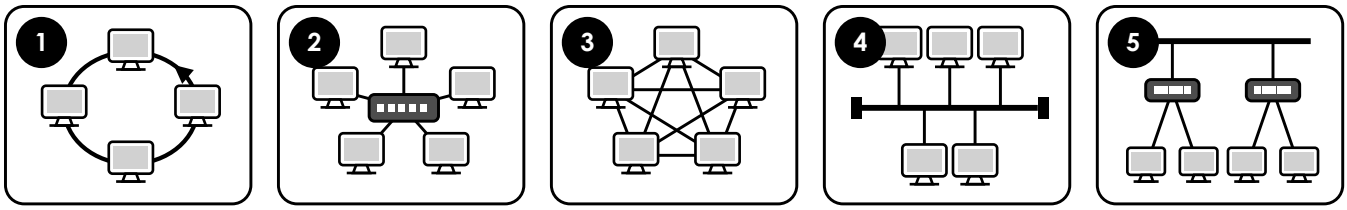
Tick one box for each network (textbook Activity, page 68).

The network	LAN	PAN	MAN	WAN
The computers of your school lab	☐	☐	☐	☐
The Internet	☐	☐	☐	☐
The branches of a bank in one city, joined together	☐	☐	☐	☐
A phone joined to earphones with Bluetooth	☐	☐	☐	☐
The computers of one office building	☐	☐	☐	☐
A university and its branches in a city	☐	☐	☐	☐

F Name the topologies

5 marks

Write the name of each topology on the line.



---------- ---------- ---------- ---------- ----------

G Benefit or disadvantage?

6 marks

Tick one box for each sentence.

Networking ...	Benefit	Disadvantage
One printer can be shared by many users.	☐	☐
A virus can spread to the other computers.	☐	☐
Hackers may try to get into the server.	☐	☐
Software can be installed on a server and shared.	☐	☐
A trained network administrator is needed.	☐	☐
Users can reach their documents any time, anywhere.	☐	☐

H Answer these

4 marks

Write full sentences.

1. Write two differences between a hub and a switch. (2 marks)

2. What is a Personal Area Network (PAN)? Give one example. (2 marks)

REVISION

Unit 2 in One Page

Read this the night before the test

Topic	The key facts
A computer	Accepts data, processes them by instructions, gives output. Speed, accuracy, efficiency, versatility, storing and retrieving.
By size	Super, mainframe (terminals), mini (banks), micro (personal). Mainframe and mini are now servers.
By technology	Analog (speed, pressure, temperature), digital (our computers), hybrid (both: the ECG).
Computer system	Input, main memory, CPU, output; secondary memory keeps data. Control signals: solid lines; data: dotted lines.
Input devices	Keyboard, mouse, touch screen (also output), joystick, light pen, webcam, scanner, barcode reader, MICR, OCR, OMR.
Output devices	Soft copy: CRT, LCD, LED, projector. Hard copy: impact printers (dot matrix, line), non impact printers (laser, ink jet, thermal), plotter. Sound: speakers.
CPU	ALU (maths and logic), Control Unit (controls devices), registers. Speed in MHz and GHz.
Primary memory	RAM (volatile, read and write), ROM (non-volatile, BIOS), cache (fast, CPU memory).
Secondary storage	Non-volatile. Magnetic (hard disk, tape), optical (CD, DVD, Blu-ray), solid state (flash drive, memory card).
Ports	PS/2, parallel (25 holes), HDMI, Ethernet RJ45, audio, USB (A, B, C), VGA (15 holes), serial.
Communication	Sender, medium, receiver. Simplex (TV), half-duplex (walkie-talkie), full duplex (phone).
Media	Guided: twisted pair (UTP, STP), coaxial, fibre optics. Unguided: radio, microwaves, infrared.
Networks	NIC, hub / switch, Wi-Fi, router, modem, firewall. LAN, PAN, MAN, WAN. Star, bus, ring, tree, mesh.

TICK WHEN YOU CAN DO IT

- [ ] I can explain what a computer is and name its five features.
- [ ] I can classify computers by size and by technology.
- [ ] I can explain the diagram of a computer system (Figure 2.10).
- [ ] I can name input and output devices and the parts of the CPU.
- [ ] I can compare RAM, ROM, cache and the kinds of secondary storage.
- [ ] I can name the ports of a computer and what goes into each.
- [ ] I can explain transmission modes and media, network devices, types and topologies.

WORD POWER

Unit 2 Word List

Twenty-nine words you must know

Word	What it means
Computer	An electronic device that accepts data, processes them by instructions and produces output.
Versatility	Being able to do many tasks at the same time.
Super computer	The most powerful kind of computer: large, expensive and rare.
Server computer	The new name for mainframe and mini computers.
Terminal	A device through which a user shares a central computer.
Hybrid computer	Uses both analog and digital technology, like the ECG machine.
Input device	A device that feeds data and instructions to the computer.
Output device	A device that gives out the processed information.
Soft / hard copy	Output on the screen / tangible output, on paper.
CPU	The Central Processing Unit, the brain of the computer: ALU, CU and registers.
ALU	The Arithmetic and Logical Unit: mathematical and logical functions.
Control Unit (CU)	The part of the CPU that controls all the devices.
RAM	Random Access Memory: the volatile main memory.
ROM	Read Only Memory: non-volatile; it keeps the BIOS.
BIOS	Basic Input Output System: the commands to boot the computer.
Cache memory	Small, fast CPU memory between the CPU and the primary memory.
Secondary storage	Storage that keeps data permanently (non-volatile).
Port	An interface that connects a computer with a device or another computer.
Data communication	Transmitting data and information.
Simplex / duplex	One way only / one way at a time (half) / both ways at once (full).
Guided media	Wired media: twisted pair, coaxial cable, fibre optics.
Unguided media	Wireless media: radio waves, microwaves, infrared.
Router	Connects two or more networks; also called a gateway.
Modem	Changes digital signals to analog and back (modulation, demodulation).
Firewall	A security barrier between the internet and a network.
Topology	The pattern of connections in a network: star, bus, ring, tree, mesh.
Plotter	An output device that draws plans with a moving pen.
MICR	Magnetic Ink Character Reader: reads the numbers on a cheque.
NIC	Network Interface Card: it joins a computer to a network.

"Learn the words today, and the ideas will follow tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 2 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. Which kind of computer has the highest computing power?
   (a) Mainframe (b) Mini (c) Super (d) Micro
2. Which of these is an output device?
   (a) Scanner (b) Plotter (c) MICR (d) Joystick
3. Which memory loses its data when the power is switched off?
   (a) ROM (b) RAM (c) Hard disk (d) DVD
4. Which part of the CPU does the mathematical and logical functions?
   (a) ALU (b) CU (c) Registers (d) Cache
5. Which port joins a computer to a network switch?
   (a) VGA (b) PS/2 (c) Ethernet (d) Parallel
6. A walkie-talkie uses the ... mode of data transmission.
   (a) simplex (b) half-duplex (c) full duplex (d) parallel
7. The fastest data transmission medium today is
   (a) UTP (b) coaxial cable (c) fibre optics (d) infrared
8. A - A modem changes digital signals to analog signals. B - A hub sends data only to the relevant computer. C - The Internet is the best example of a WAN. Which are correct?
   (a) A and B only (b) A and C only (c) B and C only (d) A, B and C

2 Fill in the blanks

8 marks

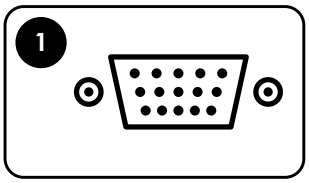
One word or number in each blank.

1. A computer that combines analog and digital technology is a ..... computer.
2. The commands to boot a computer, the BIOS, are kept in ..... .
3. The memory between the CPU and the primary memory is the ..... memory.
4. The VGA port has ..... holes.
5. Printouts on paper are called ..... copies.
6. In a ..... topology every computer is joined to every other computer.
7. The device that connects two or more networks is the ..... .
8. Wi-Fi and Bluetooth use ..... waves.

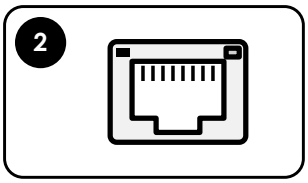
3 Look at the pictures

8 marks

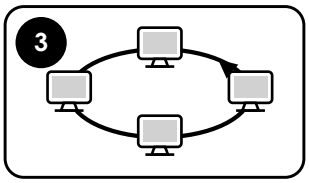
Write the name of each one, and what it is used for or how it works.

1 

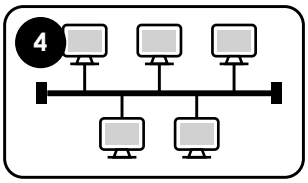
Name: .....

2 

Name: .....

3 

Name: .....

4 

Name: .....

4 Answer these

16 marks

Write in the space given.

1. Name the three main parts of the CPU and write what each part does. (3 marks)

2. Write two differences between RAM and ROM. (2 marks)

3. Name the three types of secondary storage devices. Give one example of each. (3 marks)

4. What is the difference between half-duplex mode and full duplex mode? Give one example of each. (2 marks)

5. Draw a star topology with four computers. Label the device in the centre. (3 marks)

6. Write two benefits and one disadvantage of computer networks. (3 marks)